

Problem set 4 Solution

Probabilistic classifiers

Exercise 1

1. For Example 2 in "Probabilistic classifiers" video, compute the maximum likelihood estimation for the parameter β when the observed data contains h heads and t tails.

Hint*: find β that maximizes the log likelihood function $h \log \beta + t \log(1 - \beta)$

2. Interpret the resulting estimate.

Solution

1. Differentiating $h \log \beta + t \log(1 - \beta)$ with respect to β and setting the result equal to 0, we get

$$\frac{h}{\beta} - \frac{t}{(1 - \beta)} = 0$$

$$h(1 - \beta) = t\beta$$

$$h - h\beta = t\beta$$

$$h = (h + t)\beta$$

$$\beta = \frac{h}{h + t}$$

2. The maximum likelihood estimate is nothing else but the relative frequency of observing heads!

Exercise 2

Feature x can take one of the three values: cat, dog, rabbit. Let $n(\text{cat}, c) = 9$, $n(\text{dog}, c) = 0$, and $n(\text{rabbit}, c) = 1$, where $n(a, c)$ denotes the number of training objects in class c with $x = a$.

1. Estimate the probabilities $P(x = \text{cat} \mid c)$, $P(x = \text{dog} \mid c)$, $P(x = \text{rabbit} \mid c)$.
2. Smooth the obtained probabilities using Laplace smoothing.

Solution

1. Let $N(c) = n(\text{cat}, c) + n(\text{dog}, c) + n(\text{rabbit}, c) = 10$ be the number of training objects in class c . Then the estimations are: - $P(x = \text{cat} \mid c) = \frac{n(\text{cat}, c)}{N(c)} = \frac{9}{10} = 0.9$ - $P(x = \text{dog} \mid c) = \frac{n(\text{dog}, c)}{N(c)} = \frac{0}{10} = 0$ - $P(x = \text{rabbit} \mid c) = \frac{n(\text{rabbit}, c)}{N(c)} = \frac{1}{10} = 0.1$
2. Smoothed estimations are: - $P(x = \text{cat} \mid c) = \frac{n(\text{cat}, c) + 1}{N(c) + 3} = \frac{9 + 1}{10 + 3} = \frac{10}{13}$ - $P(x = \text{dog} \mid c) = \frac{n(\text{dog}, c) + 1}{N(c) + 3} = \frac{0 + 1}{10 + 3} = \frac{1}{13}$ - $P(x = \text{rabbit} \mid c) = \frac{n(\text{rabbit}, c) + 1}{N(c) + 3} = \frac{1 + 1}{10 + 3} = \frac{2}{13}$
3. Imagine that you are given the following set of training examples. Each feature can take on one of three nominal values: a , b , or c .

F_1	F_2	F_3	Class Label
a	c	b	+
c	a	c	+
a	a	c	-
b	c	a	-
c	c	b	-

How would a Naive Bayes system classify the following test example?
 $F_1 = a, F_2 = c, F_3 = b$

$$\begin{aligned}
 P(c = + \mid F_1 = a, F_2 = c, F_3 = b) &\propto P(c = +)P(F_1 = a \mid c = +)P(F_2 = c \mid c = +)P(F_3 = b \mid c = +) \\
 &\propto \frac{2}{5} * \frac{1}{2} * \frac{1}{2} * \frac{1}{2} \\
 &\propto \frac{1}{20}
 \end{aligned}$$

$$\begin{aligned}
 P(c = - \mid F_1 = a, F_2 = c, F_3 = b) &\propto P(c = -)P(F_1 = a \mid c = -)P(F_2 = c \mid c = -)P(F_3 = b \mid c = -) \\
 &\propto \frac{3}{5} * \frac{1}{3} * \frac{2}{3} * \frac{1}{3} \\
 &\propto \frac{2}{45}
 \end{aligned}$$

Hence $P(c = + \mid F_1 = a, F_2 = c, F_3 = b) > P(c = - \mid F_1 = a, F_2 = c, F_3 = b)$

Exercise 4

1% of women at age forty who participate in routine screening have breast cancer. 80% of women with breast cancer will get positive mammographies.

9.6% of women without breast cancer will also get positive mammographies. A woman in this age group had a positive mammography in a routine screening. What is the probability that she actually has breast cancer?

Solution

Given:

$$P(C) = 0.01 \quad (\text{probability a woman has cancer})$$

$$P(\neg C) = 0.99$$

$$P(+ | C) = 0.80 \quad (\text{true positive rate})$$

$$P(+ | \neg C) = 0.096 \quad (\text{false positive rate})$$

We are asked to compute:

$$P(C | +)$$

Using Bayes' theorem:

$$P(C | +) = \frac{P(+ | C)P(C)}{P(+)}$$

First compute $P(+)$ using the law of total probability:

$$P(+)=P(+|C)P(C)+P(+|\neg C)P(\neg C)$$

$$P(+)=0.80(0.01)+0.096(0.99)$$

$$P(+)=0.008+0.09504$$

$$P(+)=0.10304$$

Now apply Bayes' theorem:

$$P(C | +) = \frac{0.80 \times 0.01}{0.10304}$$

$$P(C | +) = \frac{0.008}{0.10304}$$

$$P(C | +) \approx 0.0776$$

$$P(C | +) \approx 7.76\%$$

Final Answer:

$P(C +) \approx 0.078 \quad (\text{about } 7.8\%)$
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Excercise 5

There is a 50% chance there is both life and water on Mars, a 25% chance there is life but no water, and a 25% chance there is no life and no water. What is the probability that there is life on Mars given that there is water?

Solution

Given:

Let L = there is life on Mars, W = there is water on Mars We are given joint probabilities:

$$P(L \cap W) = 0.50, P(L \cap \neg W) = 0.25, P(\neg L \cap \neg W) = 0.25$$

(No probability is assigned to $\neg L \cap W$, so it is 0.)

We need to compute: $P(L | W)$

Using Bayes' rule (conditional probability definition): $P(L | W) = \frac{P(L \cap W)}{P(W)}$

First compute $P(W)$:

$$P(W) = P(L \cap W) + P(\neg L \cap W)$$

$$P(W) = 0.50 + 0 = 0.50$$

Now compute the conditional probability:

$$P(L | W) = \frac{0.50}{0.50} = 1$$

Final Answer:

$$P(L | W) = 1$$

If water is observed, the probability that life exists on Mars is 100%.

Excercise 6

In a medical study, 100 patients all fell into one of three classes: Pneumonia, Flu, or Healthy. The following database indicates how many patients in each class had fever and headache.

Pneumonia		
Fever	Headache	Count
T	T	5
T	F	0
F	T	4
F	F	1

Flu		
Fever	Headache	Count
T	T	9
T	F	6
F	T	3
F	F	2

Healthy		
Fever	Headache	Count
T	T	2
T	F	3
F	T	7
F	F	58

What values would a naive Bayes classifier assign to the three possible diagnoses? Show your work. (For this question, the three values need not sum to 1. Recall that the naive Bayes classifier drops the denominator because it is the same for all three classes.)

Solution

From the tables we get the class counts (summing rows within each class):

$$\begin{aligned}
 N(\text{Pneumonia}) &= 5 + 0 + 4 + 1 = 10, \\
 N(\text{Flu}) &= 9 + 6 + 3 + 2 = 20, \\
 N(\text{Healthy}) &= 2 + 3 + 7 + 58 = 70, \\
 N(\text{total}) &= 100.
 \end{aligned}$$

Thus the class priors are

$$P(\text{Pneumonia}) = \frac{10}{100} = \frac{1}{10}, \quad P(\text{Flu}) = \frac{20}{100} = \frac{1}{5}, \quad P(\text{Healthy}) = \frac{70}{100} = \frac{7}{10}.$$

Compute the conditional probabilities (within each class) for the observed features Fever=T and Headache=T.

Pneumonia:

$$P(\text{Fever} = T \mid \text{Pneumonia}) = \frac{5 + 0}{10} = \frac{5}{10} = \frac{1}{2}, \quad P(\text{Headache} = T \mid \text{Pneumonia}) = \frac{5 + 4}{10} = \frac{9}{10}.$$

Naive Bayes (features independent given class) gives the unnormalized score

$$\begin{aligned}
 \tilde{P}(\text{Pneumonia} \mid \text{data}) &= P(\text{Pneumonia}) P(F = T \mid \text{Pneumonia}) P(H = T \mid \text{Pneumonia}) \\
 &= \frac{1}{10} \cdot \frac{1}{2} \cdot \frac{9}{10} = \frac{9}{200} = 0.045.
 \end{aligned}$$

Flu:

$$P(\text{Fever} = T \mid \text{Flu}) = \frac{9 + 6}{20} = \frac{15}{20} = \frac{3}{4}, \quad P(\text{Headache} = T \mid \text{Flu}) = \frac{9 + 3}{20} = \frac{12}{20} = \frac{3}{5},$$

so

$$\tilde{P}(\text{Flu} \mid \text{data}) = \frac{1}{5} \cdot \frac{3}{4} \cdot \frac{3}{5} = \frac{9}{100} = 0.09.$$

Healthy:

$$P(\text{Fever} = T \mid \text{Healthy}) = \frac{2+3}{70} = \frac{5}{70} = \frac{1}{14}, \quad P(\text{Headache} = T \mid \text{Healthy}) = \frac{2+7}{70} = \frac{9}{70},$$

so

$$\tilde{P}(\text{Healthy} \mid \text{data}) = \frac{7}{10} \cdot \frac{1}{14} \cdot \frac{9}{70} = \frac{63}{9800} = \frac{9}{1400} \approx 0.006428571.$$

These are the (unnormalized) scores the naive Bayes classifier assigns to each class:

$\tilde{P}(\text{Pneumonia} \mid \text{data}) = \frac{9}{200} = 0.045, \quad \tilde{P}(\text{Flu} \mid \text{data}) = \frac{9}{100} = 0.09, \quad \tilde{P}(\text{Healthy} \mid \text{data}) = \frac{9}{1400} \approx 0.00643.$

Since the denominator (evidence) is common to all classes and is dropped, the classifier chooses the class with the largest unnormalized score. Here the largest score is for *Flu* (0.09), so the naive Bayes classifier predicts:

Predicted diagnosis: Flu.

Exercise 7

What is an important drawback of maximum likelihood estimation?

Solution Overfitting

Exercise 8

We want to build a naive bayes sentiment classifier using add laplace smoothing, as described in the lecture. Here is our training corpus:

- the movie has no plot
- honestly pretty boring
- + pretty interesting movie

Compute the prior for the two classes + and -, and the likelihoods for each word given the class (leave in the form of fractions).

Solution

Training corpus (labels shown):

- the movie has no plot
- honestly pretty boring
- + pretty interesting movie

Step 1: priors. There are 3 documents total: 1 positive and 2 negative, so

$$P(+) = \frac{1}{3}, \quad P(-) = \frac{2}{3}.$$

Step 2: token counts and vocabulary.

We lowercase and split on whitespace. Word counts by class:

word	count in (+)	count in (-)
the	0	1
movie	1	1
has	0	1
no	0	1
plot	0	1
honestly	0	1
pretty	1	1
boring	0	1
interesting	1	0
(total tokens)	$N_+ = 3$	$N_- = 8$

The vocabulary V is the set of unique words seen across all documents:

$$V = \{\text{the, movie, has, no, plot, honestly, pretty, boring, interesting}\}, \quad |V| = 9.$$

Step 3: add-1 (Laplace) smoothed likelihoods.

For any word w ,

$$P(w \mid c) = \frac{\text{count}(w, c) + 1}{N_c + |V|}.$$

Here $N_+ = 3$, $N_- = 8$, $|V| = 9$. Thus the denominators are

$$N_+ + |V| = 3 + 9 = 12, \quad N_- + |V| = 8 + 9 = 17.$$

We list $P(w \mid +)$ and $P(w \mid -)$ (fractions left in rational form and simplified where convenient):

word	$P(w \mid +) = \frac{\text{count}(w, +) + 1}{12}$	$P(w \mid -) = \frac{\text{count}(w, -) + 1}{17}$
the	$\frac{0+1}{12} = \frac{1}{12}$	$\frac{1+1}{17} = \frac{2}{17}$
movie	$\frac{1+1}{12} = \frac{2}{12} = \frac{1}{6}$	$\frac{1+1}{17} = \frac{2}{17}$
has	$\frac{0+1}{12} = \frac{1}{12}$	$\frac{1+1}{17} = \frac{2}{17}$
no	$\frac{0+1}{12} = \frac{1}{12}$	$\frac{1+1}{17} = \frac{2}{17}$
plot	$\frac{0+1}{12} = \frac{1}{12}$	$\frac{1+1}{17} = \frac{2}{17}$
honestly	$\frac{0+1}{12} = \frac{1}{12}$	$\frac{1+1}{17} = \frac{2}{17}$
pretty	$\frac{1+1}{12} = \frac{2}{12} = \frac{1}{6}$	$\frac{1+1}{17} = \frac{2}{17}$
boring	$\frac{0+1}{12} = \frac{1}{12}$	$\frac{1+1}{17} = \frac{2}{17}$
interesting	$\frac{1+1}{12} = \frac{2}{12} = \frac{1}{6}$	$\frac{0+1}{17} = \frac{1}{17}$

Final answers (priors and likelihoods):

$$\boxed{P(+) = \frac{1}{3}, \quad P(-) = \frac{2}{3}}$$

and for each vocabulary word w ,

$$\boxed{P(w \mid +) = \frac{\text{count}(w, +) + 1}{12}, \quad P(w \mid -) = \frac{\text{count}(w, -) + 1}{17}}$$

with the numeric fractions given in the table above (e.g. $P(\text{pretty} \mid +) = \frac{1}{6}$, $P(\text{pretty} \mid -) = \frac{2}{17}$, etc.).

Exercise 9

In machine learning classification, the *discriminative approach* models the conditional probability of the class label given the observed features, i.e.,

$$P(y \mid x),$$

rather than modeling the joint distribution $P(x, y)$ or the class-conditional likelihood $P(x \mid y)$. State whether this statement is **True** or **False**.

Solution: True

Exercise 10

Applying Naïve Bayes to data with numerical attributes and using the Laplace correction Given the training data in the table below (Tennis data with some numerical attributes), predict the class of the following new example using Naïve Bayes classification: outlook=overcast, temperature=60, humidity=62, windy=false.

outlook	temperature	humidity	windy	play
sunny	85	85	false	no
sunny	80	90	true	no
overcast	83	86	false	yes
rainy	70	96	false	yes
rainy	68	80	false	yes
rainy	65	70	true	no
overcast	64	65	true	yes
sunny	72	95	false	no
sunny	69	70	false	yes
rainy	75	80	false	yes
sunny	75	70	true	yes
overcast	72	90	true	yes
overcast	81	75	false	yes
rainy	71	91	true	no

Table 1: Play Tennis Dataset

Solution:

First, we need to calculate the mean μ and standard deviation σ values for the numerical attributes.

X_i , $i = 1 \dots n$ – the i -th measurement, n – number of measurements

$$\mu = \frac{\sum_{i=1}^n X_i}{n}$$
$$\sigma^2 = \frac{\sum_{i=1}^n (X_i - \mu)^2}{n - 1}$$

$$\mu_{\text{temp_yes}} = 73, \quad \sigma_{\text{temp_yes}} = 6.2 \quad \mu_{\text{temp_no}} = 74.6, \quad \sigma_{\text{temp_no}} = 8.0$$

$$\mu_{\text{hum_yes}} = 79.1, \quad \sigma_{\text{hum_yes}} = 10.2 \quad \mu_{\text{hum_no}} = 86.2, \quad \sigma_{\text{hum_no}} = 9.7$$

Second, to calculate $f(\text{temperature} = 60 \mid \text{yes})$, $f(\text{temperature} = 60 \mid \text{no})$, $f(\text{humidity} = 62 \mid \text{yes})$ and $f(\text{humidity} = 62 \mid \text{no})$ using the probability density function for the normal distribution:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$f(\text{temperature} = 60 \mid \text{yes}) = \frac{1}{6.2\sqrt{2\pi}} e^{-\frac{(60-73)^2}{2(6.2)^2}} = 0.071$$

$$f(\text{temperature} = 60 \mid \text{no}) = \frac{1}{8\sqrt{2\pi}} e^{-\frac{(60-74.6)^2}{2(8)^2}} = 0.0094$$

$$f(\text{humidity} = 62 \mid \text{yes}) = \frac{1}{10.2\sqrt{2\pi}} e^{-\frac{(62-79.1)^2}{2(10.2)^2}} = 0.0096$$

$$f(\text{humidity} = 62 \mid \text{no}) = \frac{1}{9.7\sqrt{2\pi}} e^{-\frac{(62-86.2)^2}{2(9.7)^2}} = 0.0018$$

Third, we can calculate the probabilities for the nominal attributes:

$$P(\text{yes}) = \frac{9}{14} = 0.643 \quad P(\text{no}) = \frac{5}{14} = 0.357$$

$$P(\text{outlook}=\text{overcast} \mid \text{yes}) = \frac{4}{14} = 0.286 \quad P(\text{outlook}=\text{overcast} \mid \text{no}) = \frac{0}{5} = 0$$

$$P(\text{windy}=\text{false} \mid \text{yes}) = \frac{6}{9} = 0.667 \quad P(\text{windy}=\text{false} \mid \text{no}) = \frac{2}{5} = 0.4$$

As $P(\text{outlook}=\text{overcast} \mid \text{no}) = 0$, we need to use a Laplace estimator for the attribute *outlook*. We assume that the three values (sunny, overcast, rainy) are equally probable.